The Expectation-Maximization (EM) Algorithm

A Comprehensive Study Guide

Based on "Statistics EM Algorithm notes by Samyak.pdf"

1. Introduction and Setup

In statistical modeling, finding the Maximum Likelihood Estimate (MLE) can be prohibitively difficult when working directly with the marginal likelihood of observed data. The **Expectation-Maximization (EM)** algorithm is an iterative technique that simplifies this optimization by introducing latent (unobserved or missing) variables.

Notation, Domains, and Terminology

To strictly define the problem, we establish the following variables and spaces:

- **Observed Data (x)**: A realization of a random variable $X \in \mathcal{X}$, where $\mathcal{X}$ is the observable space (e.g., $\mathbb{R}^n$). This is the incomplete data.
- **Latent/Missing Data (y)**: A realization of a random variable $Y \in \mathcal{Y}$, where $\mathcal{Y}$ is the unobserved latent space.
- **Complete Data**: The pair $(x, y) \in \mathcal{X} \times \mathcal{Y}$.
- **Parameter Space (Ω)**: The domain of unknown parameters, typically $\Omega \subseteq \mathbb{R}^K$.
- **Unknown Parameters (Ψ)**: The vector of parameters $\Psi \in \Omega$ that we wish to estimate.
- **Likelihood (Incomplete Data)**: $L(\Psi) = p(x|\Psi)$, representing the marginal density of X.
- **Complete Likelihood**: $L_c(\Psi; x, y) = p(x, y|\Psi)$, representing the joint density of (X, Y).
- **Complete Log-Likelihood**: $\ell_c(\Psi; x, y) = \ln p(x, y|\Psi)$.

Goal: Maximize the incomplete log-likelihood $\ell(\Psi) = \ln L(\Psi)$ with respect to $\Psi \in \Omega$. Because directly finding $\hat{\Psi}_{MLE} = \text{argmax}_\Psi \ell(\Psi)$ is often intractable, EM optimizes it indirectly via the complete log-likelihood.

Note on notational abuse: x, y are used to represent both the random variables (X, Y) and their specific realisations. Similarly, $p(\cdot)$ denotes the probability mass or density function depending on the context.

2. The EM Algorithm

The EM algorithm starts with an initial parameter guess $\Psi^{(0)} \in \Omega$ and iteratively generates a sequence of estimates $\{\Psi^{(t)}\}_{t=0}^{\infty}$. At the t-th iteration ($t = 0, 1, 2, \dots$), we assume the conditional distribution $p(y|x, \Psi^{(t)})$ can be found.

The Core Steps of the EM Algorithm

1. Expectation (E-step):
Calculate the expected value of the complete log-likelihood, where the expectation is taken with respect to the conditional distribution of Y given the observed data x and the current parameter estimate $\Psi^{(t)}$. We define this expected complete log-likelihood as the

Q-function:

$$Q(\Psi|\Psi^{(t)}) = \mathbb{E}_{Y|x,\Psi^{(t)}}[\ln p(x,y|\Psi) \mid x, \Psi^{(t)}] = \int_{\mathcal{Y}} \ln p(x,y|\Psi) p(y|x,\Psi^{(t)}) dy \tag{1}$$

(If Y is discrete, the integral is replaced by a sum).

2. Maximization (M-step):
Maximize the Q-function with respect to $\Psi \in \Omega$ to find the next parameter estimate $\Psi^{(t+1)}$:

$$\Psi^{(t+1)} = \text{argmax}_{\Psi\in\Omega}\, Q(\Psi|\Psi^{(t)}) \tag{2}$$

Stopping Criterion:
The algorithm halts when the change in the incomplete log-likelihood falls below a small threshold ϵ (e.g., 10^{-6}):

$$|\ell(\Psi^{(t+1)}) - \ell(\Psi^{(t)})| < \epsilon \implies \hat{\Psi}_{MLE} \approx \Psi^{(t+1)}$$

Generalized EM (GEM) Algorithm

If the M-step is mathematically difficult to solve for an absolute maximum, GEM suggests it is sufficient to choose *any* $\Psi^{(t+1)} \in \Omega$ that strictly improves the Q-function:

$$Q(\Psi^{(t+1)}|\Psi^{(t)}) \geq Q(\Psi^{(t)}|\Psi^{(t)})$$

3. Theoretical Properties

Monotonicity of EM

A fundamental property of EM is that it never decreases the true log-likelihood, i.e., $\ell(\Psi^{(t+1)}) \geq \ell(\Psi^{(t)})$ for all t.

Proof. Using Bayes' rule, $p(x,y|\Psi) = p(y|x,\Psi)p(x|\Psi)$. Taking the natural logarithm:

$$\ln p(x,y|\Psi) = \ln p(y|x,\Psi) + \ln p(x|\Psi)$$

Taking the expectation $\mathbb{E}_{Y|x,\Psi^{(t)}}[\cdot]$ on both sides:

$$Q(\Psi|\Psi^{(t)}) = H(\Psi|\Psi^{(t)}) + \ell(\Psi)$$

where $H(\Psi|\Psi^{(t)}) = \mathbb{E}[\ln p(y|x,\Psi)|x,\Psi^{(t)}]$. Rearranging gives $\ell(\Psi) = Q(\Psi|\Psi^{(t)}) - H(\Psi|\Psi^{(t)})$.

To show $\ell(\Psi^{(t+1)}) \geq \ell(\Psi^{(t)})$, we know from the M-step that $Q(\Psi^{(t+1)}|\Psi^{(t)}) \geq Q(\Psi^{(t)}|\Psi^{(t)})$. It remains to show that $H(\Psi^{(t+1)}|\Psi^{(t)}) \leq H(\Psi^{(t)}|\Psi^{(t)})$:

$$\begin{aligned}
H(\Psi^{(t+1)}|\Psi^{(t)}) - H(\Psi^{(t)}|\Psi^{(t)}) &= \mathbb{E}\left[\ln \frac{p(y|x,\Psi^{(t+1)})}{p(y|x,\Psi^{(t)})} \,\middle|\, x, \Psi^{(t)}\right] \\
&\leq \ln\left(\mathbb{E}\left[\frac{p(y|x,\Psi^{(t+1)})}{p(y|x,\Psi^{(t)})} \,\middle|\, x, \Psi^{(t)}\right]\right) \quad \text{(by Jensen's Inequality)} \\
&= \ln\left(\int_{\mathcal{Y}} \frac{p(y|x,\Psi^{(t+1)})}{p(y|x,\Psi^{(t)})} p(y|x,\Psi^{(t)}) dy\right) \\
&= \ln\left(\int_{\mathcal{Y}} p(y|x,\Psi^{(t+1)}) dy\right) = \ln(1) = 0.
\end{aligned}$$

Thus, $\ell(\Psi^{(t+1)}) \geq \ell(\Psi^{(t)})$, proving monotonic convergence (assuming L is bounded). □

Convergence (Wu 1983) and MM Connection

Suppose $Q(\Psi|\phi)$ is continuous in both Ψ and $\phi \in \Omega$. Then all limit points of $\{\Psi^{(t)}\}_0^\infty$ are stationary points of $L(\Psi)$ (i.e., $\frac{\partial}{\partial\Psi}L(\Psi) = 0$). The sequence $L(\Psi^{(t)})$ converges monotonically to some $L(\Psi^*)$. If $L(\Psi)$ is unimodal with a unique stationary point Ψ^* and $\frac{\partial}{\partial\Psi}Q(\Psi|\phi)$ is continuous, then $\Psi^{(t)}$ converges to the true MLE. *Note: EM is a special case of the broader MM (Minorize-Maximization) algorithmic framework.*

4. Application: Finite Mixture Distributions

Let the observed data be $x = (x_1, \ldots, x_n)$ i.i.d., sampled from a mixture of K probability distributions.

$$p(x_i|\Psi) = \sum_{j=1}^{K} \pi_j f(x_i|\theta_j)$$

- $f(x|\theta_j)$: PDF of the j-th class. Let's assume a Gaussian Mixture Model (GMM), where $f(x|\theta_j) = \frac{1}{\sigma_j\sqrt{2\pi}} e^{-\frac{(x-\mu_j)^2}{2\sigma_j^2}}$.
- Parameters: $\Psi = (\theta_1, \ldots, \theta_K, \pi_1, \ldots, \pi_K)$, where $\theta_j = (\mu_j, \sigma_j^2)$.
- Constraint: $\sum_{j=1}^{K} \pi_j = 1,\ \pi_j \geq 0$.

Latent Variables: Introduce $y_i = (y_{i1}, \ldots, y_{iK}) \sim \text{Multinomial}(1, \pi_1, \ldots, \pi_K)$, where $y_{ij} = 1$ if x_i comes from the j-th class (i.e., $Y_i = e_j$), and 0 otherwise. The complete data is (x, y), and the complete likelihood is:

$$p(x, y|\Psi) = \prod_{i=1}^{n} \prod_{j=1}^{K} [\pi_j f(x_i|\theta_j)]^{y_{ij}}$$

$$\ell_c(\Psi) = \sum_{i=1}^{n} \sum_{j=1}^{K} y_{ij} \{\ln \pi_j + \ln f(x_i|\theta_j)\}$$

EM Algorithm for GMMs

E-step: Compute $Q(\Psi|\Psi^{(t)})$. The only missing terms are the latent variables y_{ij}. We calculate their expectation given the data and current parameters, which we denote as $w_{ij}^{(t)}$:

$$w_{ij}^{(t)} = P(y_{ij} = 1|x_i, \Psi^{(t)}) = \frac{P(y_i = e_j|\Psi^{(t)})P(x_i|y_i = e_j, \Psi^{(t)})}{P(x_i|\Psi^{(t)})} = \frac{\pi_j^{(t)} f(x_i|\theta_j^{(t)})}{\sum_{l=1}^{K} \pi_l^{(t)} f(x_i|\theta_l^{(t)})} \quad (3)$$

This acts as the posterior probability that x_i belongs to class j. The Q-function becomes:

$$Q(\Psi|\Psi^{(t)}) = \sum_{i=1}^{n} \sum_{j=1}^{K} w_{ij}^{(t)} \ln \pi_j + \sum_{i=1}^{n} \sum_{j=1}^{K} w_{ij}^{(t)} \ln f(x_i|\theta_j)$$

M-step: Maximize $Q(\Psi|\Psi^{(t)})$ with respect to π_j, μ_j, σ_j^2.

- **For π_j:** We use a Lagrange multiplier λ to enforce $\sum_{j=1}^{K} \pi_j = 1$. Let $R(\pi) = Q - \lambda(\sum \pi_j - 1)$.

$$\frac{\partial R}{\partial \pi_j} = \sum_{i=1}^{n} \frac{w_{ij}^{(t)}}{\pi_j} - \lambda = 0 \implies \lambda \pi_j = \sum_{i=1}^{n} w_{ij}^{(t)}$$

Summing over j, we get $\lambda \sum_j \pi_j = \sum_i \sum_j w_{ij}^{(t)} \implies \lambda = n$. Thus:

$$\pi_j^{(t+1)} = \frac{1}{n} \sum_{i=1}^{n} w_{ij}^{(t)}$$

- **For μ_j:** Substitute $f(x_i|\theta_j)$ and set derivative w.r.t μ_j to 0.

$$\frac{\partial}{\partial \mu_j} \sum_{i,j} w_{ij}^{(t)} \left(-\frac{(x_i - \mu_j)^2}{2\sigma_j^2} \right) = 0 \implies \sum_{i=1}^{n} w_{ij}^{(t)} (x_i - \mu_j) = 0 \implies \mu_j^{(t+1)} = \frac{\sum_{i=1}^{n} w_{ij}^{(t)} x_i}{\sum_{i=1}^{n} w_{ij}^{(t)}}$$

- **For σ_j^2:** Take the derivative w.r.t σ_j^2 (treating it as a single variable $v_j = \sigma_j^2$).

$$\frac{\partial}{\partial \sigma_j^2} \sum_{i=1}^{n} w_{ij}^{(t)} \left(-\frac{1}{2} \ln \sigma_j^2 - \frac{(x_i - \mu_j^{(t+1)})^2}{2\sigma_j^2} \right) = 0 \implies (\sigma_j^2)^{(t+1)} = \frac{\sum_{i=1}^{n} w_{ij}^{(t)} (x_i - \mu_j^{(t+1)})^2}{\sum_{i=1}^{n} w_{ij}^{(t)}}$$

5. Application: Dempster, Laird, Rubin (1977) Example

Consider 197 animals categorized into 4 classes: A, B, C, D. The observed counts are $x = (x_1, x_2, x_3, x_4)$, following a Multinomial distribution:

$$x \sim \text{Multinomial}\left(n = 197, \frac{1}{2} + \frac{\theta}{4}, \frac{1-\theta}{4}, \frac{1-\theta}{4}, \frac{\theta}{4} \right)$$

The incomplete log-likelihood is $\ell(\theta) = C + x_1 \ln\left(\frac{1}{2} + \frac{\theta}{4}\right) + (x_2 + x_3)\ln(1-\theta) + x_4 \ln\theta$. Direct maximization requires solving a messy polynomial.

Introducing Latent Variable: Divide category A into two latent categories A_1 and A_2, such that $x_1 = x_{11} + x_{12}$. We assign probabilities $1/2$ and $\theta/4$ to them, respectively. Our missing data is $y = x_{12}$. The complete data $(x_{11}, x_{12}, x_2, x_3, x_4)$ follows:

$$\text{Multinomial}\left(n, \frac{1}{2}, \frac{\theta}{4}, \frac{1-\theta}{4}, \frac{1-\theta}{4}, \frac{\theta}{4} \right)$$

The complete log-likelihood simplifies remarkably:

$$\ell_c(\theta) = C' + (x_{12} + x_4)\ln\theta + (x_2 + x_3)\ln(1-\theta)$$

E-step: We need $Q(\theta|\theta^{(t)}) = \mathbb{E}[\ell_c(\theta)|x, \theta^{(t)}]$. The only missing term is x_{12}. Given x_1 and $\theta^{(t)}$, x_{12} is Binomially distributed:

$$x_{12} \mid x_1, \theta^{(t)} \sim \text{Binomial}\left(x_1, \frac{\theta^{(t)}/4}{1/2 + \theta^{(t)}/4} \right)$$

Its expectation is $q_t = \mathbb{E}[x_{12}|x_1, \theta^{(t)}] = x_1 \frac{\theta^{(t)}}{2+\theta^{(t)}}$. Substituting this into the complete log-likelihood gives:

$$Q(\theta|\theta^{(t)}) = C' + (q_t + x_4)\ln\theta + (x_2 + x_3)\ln(1-\theta)$$

M-step: Maximize Q with respect to θ:

$$\frac{\partial Q}{\partial \theta} = \frac{q_t + x_4}{\theta} - \frac{x_2 + x_3}{1-\theta} = 0$$

Rearranging $\frac{1-\theta}{\theta} = \frac{x_2+x_3}{q_t+x_4}$ gives the parameter update (noting that $n = x_1 + x_2 + x_3 + x_4$):

$$\theta^{(t+1)} = \frac{q_t + x_4}{(n - x_1) + q_t + x_4} = \frac{x_1 \frac{\theta^{(t)}}{2+\theta^{(t)}} + x_4}{(x_2 + x_3) + x_1 \frac{\theta^{(t)}}{2+\theta^{(t)}} + x_4}$$

We iterate until $|\theta^{(t+1)} - \theta^{(t)}| < \epsilon$ or $\left|\frac{\theta^{(t+1)} - \theta^{(t)}}{\theta^{(t)}}\right| < \epsilon$, converging to $\hat{\theta}_{MLE}$.